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INFORMATION PROCESSING DEVICE AND INFORMATION PROCESSING METHOD

Abstract

An information processing device includes a processor configured to write a computer program, which is read from an external memory, in an internal memory, and execute the computer program read from the internal memory. The processor is configured to calculate the parity value of an instruction or a set of data of the computer program read from the external memory, and write the instruction or the set of data and the parity value in the internal memory. The processor is configured to determine, based on the parity value from the internal memory, the presence or absence of an error in the instruction or the set of data. The processor is configured to transfer, when the error has occurred, the instruction or the set of data, which is read again from the external memory, to the first writing and the second writing.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority to and incorporates by reference the entire contents of Japanese Patent Application No. 2024-030513 filed in Japan on Feb. 29, 2024.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present invention relates to an information processing device and an information processing method.

2. Description of the Related Art

[0003] A conventional information processing device includes an integrated circuit (IC) and includes an external read only memory (external ROM) that is used to store information such as computer programs to be executed in the IC. Usually, as compared to accessing the computer programs stored in the external ROM, the IC can make high-speed access to the computer programs stored in the internal random access memory (internal RAM) thereof, and hence can perform high-speed processing of those computer programs.

[0004] Generally, in a RAM in which computer programs are stored, if bit inversion such as inversion of a single bit of a stored computer program occurs due to some reason, it is believed that the computer program is not executed in a normal manner. Particularly, in a system required to perform stable operations, such as in the control system of a plant; abnormal operations attributed to bit inversion have a significant impact.

[0005] As the cause for bit inversion, for example, it is possible to think of some hardware that is persistently malfunctioning or to think of a soft error that is temporary by nature. A hardware error is caused by physical factors, and represents physical malfunctioning attributed to malfunctioning due to aging or overvoltage. Hence, a hardware error cannot be repaired by resetting the power supply. On the other hand, a soft error is caused by an external factor such as the neutron rays, which are included in radiation, or the electromagnetic noise. Thus, when a soft error occurs, since the hardware is in good health, the soft error can be repaired by restoring the correct values by means of resetting the power supply.

[0006] In recent years, from the perspective of achieving performance upgrade and energy conservation in devices, there is a demand for miniaturization and power saving of semiconductors. However, such miniaturization and power saving leads to a decrease in the resistance characteristic of the semiconductors with respect to the external influences such as radiation and electromagnetic noise. That results in an increase in the occurrence rate of soft errors.

[0007] Japanese Patent Application Laid-open No. 2007-18414

[0008] Regarding the conventional information processing devices, it is a fact that there is a demand for an information processing device in which, even if a soft error occurs in a computer program, it becomes possible to resolve the soft error and to continue with the normal operation of the computer program.

[0009] As an aspect, it is an objective to provide an information processing device that is capable of continuing with the normal operation of a computer program.

SUMMARY OF THE INVENTION

[0010] According to an aspect of an embodiment, an information processing device includes a processor configured to write a computer program, which is read from an external memory, in an internal memory and execute the computer program which is read from the internal memory. The processor is configured to calculate a parity value of an instruction or a set of data of the computer program read from the external memory, and write the instruction or the set of data and the parity value in the internal memory. The processor is configured to determine, based on the parity value of the instruction or the set of data of the computer program read from the internal memory, presence or absence of an error in the instruction or the set of data of the computer program. The processor is configured to transfer, when the error has occurred, the instruction or the set of data of the computer program, which is read again from the external memory, to the first writing of the computer program and the second writing of the instruction or the set.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. **1** is a block diagram illustrating an example of an information processing device according to an embodiment;

[0012] FIG. **2** is a block diagram illustrating an example of a memory control unit;

[0013] FIG. **3** is a flowchart for explaining an example of the operations performed in a writing operation in the information processing device;

[0014] FIG. **4** is a flowchart for explaining an example of the operations performed in an execution operation in the information processing device; and

[0015] FIG. **5** is a block diagram illustrating an example of an information processing device according to a comparison example.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

Comparison Example

[0016] FIG. **5** is a block diagram illustrating an example of an information processing device **100** according to a comparison example. The information processing device **100** illustrated in FIG. **5** includes an integrated circuit (IC) **101** and an external read only memory (external ROM) **102**. The external ROM **102** is an external memory unit used to store information such as the computer programs to be executed in the IC **101**. The IC **101** is a control unit that controls the information processing device **100** in entirety. The IC **101** reads the computer programs stored in the external ROM **102**, and executes them as functions.

[0017] The IC **101** includes an external bus interface (I/F) **111**, a random access memory (RAM) **112**, a central processing unit (CPU) **113**, and an internal bus **114**. The external bus I/F **111** represents an input-output interface for establishing connection with the external ROM **102**. The RAM **112** is an internal memory unit used to temporarily store the computer programs that have been read from the external ROM **102**. The CPU **113** controls the IC **101** in entirety. The CPU **113** writes, in the RAM **112**, the computer programs that have been read from the external ROM **102** via the external bus I/F **111**. The CPU **113** reads instructions (sets of data) of the computer programs from the RAM **112** and executes those instructions (sets of data). Herein, for explanatory convenience, an instruction (set of data) of a computer program is assumed to imply either an instruction or a set of data of the computer program. The internal bus **114** represents a bus line that establishes connection among the external bus I/F **111**, the RAM **112**, and the CPU **113**, thereby enabling communication of information such as data or computer programs among the external bus I/F **111**, the RAM **112**, and the CPU **113**.

[0018] For example, at the time of switching on the power supply to the information processing device **100**, the CPU **113** reads computer programs stored in the external ROM **102**, transfers those computer programs to the RAM **112**, and writes them in the RAM **112**. Then, for example, at the time of running a computer program, the CPU **113** reads the instructions (sets of data) of that computer program written in the RAM **112**, and executes the read instructions (sets of data).

[0019] In the conventional information processing device **100**, usually, as compared to the computer programs stored in the external ROM **102**, the computer programs stored in the internal RAM **112** can be accessed with high-speed, thereby enabling high-speed processing of those computer programs.

[0020] As a measure to deal with the soft errors in the information processing device **100**, for example, it is possible to think of installing, in the information processing device **100**, an error detection/correction function based on the error correction code (ECC).

[0021] However, regarding the ECC, more errors can be detected/corrected if the length of the error correction code is increased. However, that not only leads to an increase in the volume of information required for calculating and detecting the code, but there also occurs an increase in the volume and the band (overhead) unnecessarily spent on the error correction code at the time of recording and transfer. As a result, in an information processing device in which the ECC is installed, it becomes necessary to provide a memory for storing the error correction code and to provide a control circuit for detecting and correcting the errors. Particularly, in order to increase the memory capacity for storing the error correction code, it becomes necessary to increase the memory capacity of the RAM. That leads to an increase in the mounting area and an increase in the circuit size of the information processing device.

[0022] Moreover, ECC-based correction does not include correction of the memory content and includes correction of only the output data. Thus, in a control system of a plant that cannot be rebooted for a long period of time, errors get accumulated and, when a plurality of errors (bit inversions) occurs in a computer program that is read, it is also possible to think that correction is no more performed.

[0023] In that regard, there is a demand for an information processing device in which, even when a soft error occurs in a computer program, it is possible to resolve the soft error without installing the ECC and to continue with the normal operation of the computer program.

[0024] An exemplary embodiment of an information processing device according to the application concerned is described below in detail with reference to the accompanying drawings. However, the technology disclosed herein is not limited by the embodiment described below. Moreover, embodiments can be appropriately modified without causing any contradictions.

Embodiment

[0025] FIG. **1** is a block diagram illustrating an example of an information processing device **1** according to the present embodiment. The information processing device **1** illustrated in FIG. **1** includes an integrated circuit (IC) **2** and an external read only memory (external ROM) **3**. For example, the information processing device **1** is a distributed control system (DCS), or a safety instrumented system (SIS), or some other factory automation (FA) device, or a field device such as a pressure transmitter, a temperature transmitter, a multi-variable transmitter, or a flowmeter. The IC **2** is, for example, an integrated circuit installed inside a DCS, an SIS, an FA device, or a field device. The IC **2** controls the information processing device **1** in entirety. The external ROM **3** is an external memory unit used to store the computer programs to be executed in the IC **2**.

[0026] The IC **2** includes an external bus I/F **4**, a memory control unit **5**, a central processing unit (CPU) **6**, a first bus **7**, and a second bus **8**. The external bus I/F **4** is an input-output interface that establishes connection with the external ROM **3**. The memory control unit **5** has a random access memory (RAM) **12** built-in as the internal memory unit, and controls the RAM **12**. The CPU **6** represents a control unit that controls the IC **2** in entirety.

[0027] The first bus **7** is, for example, an internal bus that connects the CPU **6** and the external bus I/F **4** to each other and connects the CPU **6** and the memory control unit **5** to each other, and that enables communication of a variety of information such as computer programs and data. The second bus **8** is, for example, an internal bus that connects the external bus I/F **4** and the memory control unit **5** to each other and that enables communication of a variety of information such as computer programs and data.

Configuration of Memory Control Unit

[0028] FIG. **2** is a block diagram illustrating an example of the memory control unit **5**. The memory control unit **5** illustrated in FIG. **2** includes a writing control unit **11**, a RAM **12**, a reading control unit **13**, and a transferring unit **14**. The writing control unit **11** performs writing control of the RAM **12**. The writing control unit **11** sequentially writes the instructions or the sets of data of a computer program read from the external ROM **3**. At the time of sequentially writing the instructions or the sets of data of a computer program read from the external ROM **3**, the writing control unit **11** calculates a first-type parity value of each instruction (set of data) of the computer program, and sequentially writes the calculated first-type parity values and the instructions (sets of data) in a corresponding manner in the RAM **12**. The first-type parity value is, for example, an even parity value or an odd parity value.

[0029] The reading control unit **13** performs reading control of the RAM **12**. In the case of reading an instruction (set of data) and the corresponding first-type parity value from the RAM **12**, the reading control unit **13** calculates a second-type parity value of the instruction (set of data) that has been read. The second-type parity value too is, for example, an even parity value or an odd parity value. The reading control unit **13** compares the calculated second-type parity value with the first-type parity value that is read from the RAM **12**, and determines whether or not the first-type parity value is same as the second-type parity value. Based on the determination result about whether or not the first-type parity value is same as the second-type parity value, the reading control unit **13** determines the presence or absence of a parity error in the instruction (set of data). For example, a parity error represents the bit inversion caused by a soft error. If the first-type parity value is same as the second-type parity value, the reading control unit **13** determines that no parity error has occurred in the instruction (set of data) read from the RAM **12**. On the other hand, when the first-type parity value is different than the second-type parity value, the reading control unit **13** determines that a parity error has occurred in the instruction (set of data) read from the RAM **12**.

[0030] When a parity error occurs in an instruction (set of data), the transferring unit **14** transfers, to the CPU **6**, an instruction (set of data) that is again read from the external ROM via the second bus **8**. On the other hand, when no parity error has occurred in the instruction (set of data), the transferring unit **14** transfers, to the CPU **6**, the instruction (set of data) that was read from the RAM **12**. The transferring unit **14** includes an output unit **15** and a reloading unit **16**. The output unit **15** is a multiplexer that switches between transferring, to the first bus **7**, the instruction (set of data) coming from the external ROM **3** and transferred from the reloading unit **16**, and transferring, to the first bus **7**, the instruction (set of data) coming from the RAM **12** and transferred from the reading control unit **13**. When a parity error has occurred in an instruction (set of data); the output unit **15** transfers, to the first bus **7**, the instruction (set of data) coming from the external ROM **3** and transferred from the reloading unit **16**. On the other hand, when no parity error has occurred in an instruction (set of data); the output unit **15** transfers, to the first bus **7**, the instruction (set of data) coming from the RAM **12** and transferred from the reading control unit **13**.

[0031] When a parity error has occurred in an instruction (set of data), the reloading unit **16** again reads the instruction (set of data) from the external ROM **3** via the second bus **8**. Then, while transferring to the output unit **15** the instruction (set of data) that is again read from the external ROM **3** the reloading unit **16**, the reloading unit **16** transfers that instruction (set of data) to the writing control unit **11** too.

[0032] When no parity error has occurred in an instruction (set of data), the reading control unit **13** transfers the instruction (set of data) to the output unit **15** of the transferring unit **14**, and performs switching control of the output unit **15** in order to establish connection between the reading control unit **13** and the first bus **7**.

[0033] When a parity error has occurred in an instruction (set of data); the reading control unit **13** outputs, to the reloading unit **16**, a control signal for the purpose of again reading the instruction (set of data) from the external ROM **3**, and performs switching control of the output unit **15** to establish connection between the reloading unit **16** and the first bus **7**. Then, the reloading unit **16** transfers, to the output unit **15** and the writing control unit **11**, the instruction (set of data) that is again read from the external ROM **3** via the second bus **8**.

[0034] Then, as explained earlier, when a parity error has occurred in an instruction (set of data), the output unit **15** transfers, to the CPU **6** via the first bus **7**, the instruction (set of data) that is again read from the reloading unit **16**. At the time of receiving the instruction (set of data) transferred from the output unit **15** via the first bus **7**; if the instruction is received, the CPU **6** executes that instruction, and, if the data is received, the CPU **6** uses that data.

[0035] Moreover, the writing control unit **11** calculates the first-type parity value of the instruction (set of data) transferred from the reloading unit **16** and, writes that instruction (set of data) and the first-type parity value in the RAM **12**.

Writing Operation

[0036] FIG. **3** is a flowchart for explaining an example of the operations performed in a writing operation in the information processing device **1**. The information processing device **1** determines whether or not the transfer timing of a transfer from the external ROM **3** to the RAM **12** is detected (Step S**11**). When the transfer timing from the external ROM **3** to the RAM **12** is detected (Yes at Step S**11**), the information processing device **1** reads an instruction (set of data) of a computer program from the external ROM **3** (Step S**12**). The external bus I/F **4** in the information processing device **1** transfers the instruction (set of data), which is read from the external ROM **3**, to the writing control unit **11** in the memory control unit **5** via the first bus **7** (Step S**13**).

[0037] The writing control unit **11** calculates the first-type parity value of the instruction (set of data) transferred thereto via the first bus **7** (Step S**14**). Then, the writing control unit **11** writes the instruction (set of data), which is transferred thereto, and the calculated parity value in the RAM **12** of the memory control unit **5** (Step S**15**). Moreover, after writing the instruction (set of data), which is transferred thereto, in the RAM **12**, the writing control unit **11** determines whether or not the written instruction (set of data) is the last instruction (last set of data) (Step S**16**). The last instruction implies the instruction that is given at the end of, for example, a plurality of instructions given in the computer program written in the RAM **12**. The last set of data implies the set of data placed at the end of, for example, a plurality of sets of data present in the computer program that is written in the RAM **12**.

[0038] When the written instruction (set of data) is the last instruction (last set of data) of the computer program (Yes at Step S**16**), the writing control unit **11** determines that no more instructions (sets of data) are left in the computer program and thus ends the operations illustrated in FIG. **3**. On the other hand, when the written instruction (set of data) is not the last instruction (not the last set of data) of the computer program (No at Step S**16**), the writing control unit **11** determines that there are more instructions (sets of data) in the computer program, and the system control returns to Step S**12** for reading another instruction (set of data) from the external ROM **3**.

[0039] Meanwhile, when the transfer timing from the external ROM **3** to the RAM **12** is not detected (Yes at Step S**11**), the information processing device **1** ends the operations illustrated in FIG. **3**.

[0040] When the transfer timing from the external ROM **3** to the RAM **12** is detected, the memory control unit **5** of the information processing device **1** calculates the first-type parity value of the instruction (set of data) read from the external ROM **3** via the first bus **7**, and writes the calculated first-type parity value and the instruction (set of data) in the RAM **12**. As a result, the information processing device **1** prepares for making high-speed access to the computer program at the time of running the computer program.

Execution Operation

[0041] FIG. **4** is a flowchart for explaining an example of the operations performed in an execution operation in the information processing device **1**. With reference to FIG. **4**, the information processing device **1** determines whether or not the running of a computer program is detected (Step S**21**). When the running of a computer program is detected (Yes at Step S**21**), the reading control unit **13** of the information processing device **1** reads an instruction (set of data) and the first-type parity value from the RAM **12** (Step S**22**).

[0042] The reading control unit **13** calculates the second-type parity value of the instruction (set of data) read from the RAM **12** (Step S**23**). The reading control unit **13** compares the first-type parity value, which is read at Step S**22**, with the second-type parity value, which is calculated at Step S**23** (Step S**24**). The reading control unit **13** determines whether or not the first-type parity value is same as the second-type parity value (Step S**25**).

[0043] When the first-type parity value is same as the second-type parity value (Yes at Step S**25**), the reading control unit **13** determines that no parity error has occurred in the read instruction (set of data) (Step S**26**). Then, in order to switch to transferring, to the first bus **7**, the instruction (set of data) that is read from the RAM **12**; the reading control unit **13** controls the output unit **15** (Step S**27**).

[0044] The output unit **15** transfers, to the CPU **6** via the first bus **7**, the instruction (set of data) read from the reading control unit **13** (Step S**28**). When the instruction is transferred at Step S**28**, the CPU **6** executes the instruction read from the RAM **12**; or, when the data is transferred at Step S**28**, the CPU **6** uses the data read from the RAM **12** (Step S**29**). Then, the CPU **6** determines whether or not the computer program has ended (Step S**40**). If the computer program has not ended (No at Step S**40**), the system control returns to the operation at Step S**22** so that the CPU **6** can read another instruction (set of data) in the computer program and the corresponding first-type parity value.

[0045] Meanwhile, when the program has ended (Yes at Step S**40**), the CPU **6** ends the operations illustrated in FIG. **4**. Thus, by executing all instructions or by using all of the data in the computer program read from the RAM **12**, the CPU **6** becomes able to make high-speed access to the computer program.

[0046] When the first-type parity value is not same as the second-type parity value (No at Step S**25**), the reading control unit **13** determines that a parity error has occurred in the instruction (set of data) read from the RAM **12** (Step S**30**). When it is determined that a parity error has occurred, the reloading unit **16** again reads the instruction (set of data) from the external ROM **3** via the second bus **8** (Step S**31**).

[0047] Meanwhile, when it is determined at Step S**30** that a parity error has occurred, the reading control unit **13** controls the output unit **15** in order to switch to transferring, to the first bus **7**, the instruction (set of data) that is again read from the reloading unit **16** (Step S**32**). Moreover, the output unit **15** transfers the instruction (set of data), which is again read from the external ROM **3**, to the CPU **6** via the first bus **7** (Step S**33**). When the instruction that is again read from the reloading unit **16** is transferred at Step S**33**, the CPU **6** executes the instruction read from the external ROM **3**; or, when the data is transferred at Step S**33**, the CPU **6** uses the data read from the external ROM **3** (Step S**34**). Then, the system control proceeds to Step S**40** for determining whether or not the computer program has ended. As a result, even when an error occurs in an instruction or in a set of data in a computer program stored in the RAM **12**, the CPU **6** can use the correct instruction or the correct set of data read from the external ROM **3** and can execute the computer program in a normal manner.

[0048] Meanwhile, when it is determined at Step S**30** that an error has occurred, while transferring, to the CPU **6** via the first bus **7**, the instruction (set of data) that is read again; the reloading unit **16** transfers the instruction (set of data), which is read again, to the writing control unit **11** too (Step S**35**). Moreover, the writing control unit **11** calculates the first-type parity value of the instruction (set of data) transferred at Step S**35** (Step S**36**) and writes the transferred instruction (set of data) and the calculated first-type parity value in the RAM **12** (Step S**37**); and the system control proceeds to Step S**40** for determining whether or not the computer program has ended. As a result, the information processing device **1** becomes able to prepare for making high-speed access at the time of reading the next instruction (set of data).

[0049] Meanwhile, when it is determined that the running of a computer program is not detected (No at Step S**21**), the reading control unit **13** ends the operations illustrated in FIG. **4**.

Effects of Embodiment

[0050] When a parity error has occurred in an instruction (set of data) in a computer program that was transferred from the external ROM **3** and that is stored in the RAM **12**, the information processing device **1** again reads the normal instruction (set of data) from the external ROM **3**.

Then, the information processing device **1** causes the CPU **6** to execute the normal instruction (set of data) that is read again. As a result, even when a parity error has occurred in the instruction (set of data) of the RAM **12**, it becomes possible to continue with the normal operation of the computer program. Moreover, without installing the ECC and by reducing the scale attributed to the addition of a circuit for detection and correction of soft errors, it becomes possible to hold down the mounting area of the circuit and at the same time to repair the bit inversion caused by the soft errors repaired and to continue with the normal operation of the computer program.

[0051] Based on the parity value of an instruction (set of data) read from the RAM **12**, the information processing device **1** determines the presence or absence of a parity error in the instruction (set of data); and, if a parity error has occurred, transfers to the CPU **6**, the instruction (set of data) that is again read from the external ROM **3**. As a result, even when a soft error occurs, it becomes possible to continue with the normal operation of the computer program.

[0052] When no parity error has occurred, the information processing device **1** transfers, to the CPU **6**, the instruction (set of data) read from the RAM **12**. As a result, it becomes possible to make high-speed access to the computer program.

[0053] When a parity error occurs, the reloading unit **16** again reads the instruction (set of data) from the external ROM **3** and transfers the instruction (set of data), which is read again, to the output unit **15** and the writing control unit **11**. Moreover, when a parity error has occurred, the output unit **15** transfers, to the CPU **6**, the instruction (set of data) that is again read by the reloading unit **16**; and, when no parity error has occurred, the output unit **15** transfers, to the CPU **6**, the instruction (set of data) read from the RAM **12**. As a result, even when a soft error occurs, it becomes possible to continue with the normal operation of the computer program.

[0054] The information processing device **1** includes the first bus **7** that connects the transferring unit **14** and the CPU **6** to each other, and includes the second bus **8** that connects the external ROM **3** and the transferring unit **14** to each other. When a parity error occurs, the reloading unit **16** again reads the instruction (set of data) from the external ROM **3** via the second bus **8**. When a parity error has occurred, the output unit **15** transfers, to the CPU **6** via the first bus **7**, the instruction that is again read by the reloading unit **16**. However, when no parity error has occurred, the output unit **15** transfers, to the CPU **6** via the first bus **7**, the instruction (set of data) that is read from the RAM **12**. Hence, even when the CPU **6** is occupying the first bus **7**, the instruction (set of data) can be read from the external ROM **3** using the second bus **8**.

[0055] The output unit **15** transfers, to the CPU **6** via the first bus **7**, the instruction (set of data) that is again read by the reloading unit **16**. Moreover, the reloading unit **16** transfers the instruction (set of data), which is read again, to the writing control unit **11**. Furthermore, the writing control unit **11** calculates the first-type parity value of the computer program that is again read by the reloading unit **16**. Moreover, the writing control unit **11** writes the instruction (set of data), which is read again, and the calculated first-type parity value in the RAM **12**. As a result, the information processing device **1** becomes able to prepare for making high-speed access at the time of reading the next instruction (set of data).

[0056] Herein, the explanation is given for the example in which, when a parity error occurs, the reloading unit **16** again reads the instruction (set of data) from the external ROM **3** via the second bus **8**, and transfers that instruction (set of data) to the CPU **6** and the writing control unit **11** via the first bus **7**. However, that is not the only possible case. Alternatively, when a parity error occurs, the reloading unit **16** again reads the instruction (set of data) from the external ROM **3** via the second bus **8**, and transfers that instruction (set of data) to the writing control unit **11**. Then, the writing control unit **11** performs the writing operation for writing the first-type parity value of the instruction (set of data) along with the instruction (set of data) in the RAM **12**. Subsequently, the reading control unit **13** performs the operation at Step S**22** illustrated in FIG. **4**. As a result, even when a soft error occurs, it becomes possible to continue with the normal operation of the computer program.

[0057] In the information processing device **1** according to the present embodiment, the explanation is given for the case in which, in addition to using the first bus **7**, the second bus **8** is also used that establishes connection with the reloading unit **16** for error correction purposes; so that, even when the CPU **6** is occupying the first bus **7**, the instruction (set of data) is read from the external ROM **3** via the second bus **8**. However, the configuration is not limited to using two buses. Alternatively, for example, even if the second bus **8** is not provided, the first bus **7** can be connected to the reloading unit **16**.
[0058] In that case, in the information processing device **1**, while the CPU **6** is reading an instruction (set of data) of a computer program using the first bus **7**, if an error is detected in the instruction (set of data) based on the parity value, the first bus **7** is released. After the first bus **7** is released, the writing control unit **11** uses the first bus **7** and writes, in the RAM **12**, the instruction (set of data) that is again read from the external ROM **3**. After the writing of the instruction (set of data) is completed, the reading control unit **13** releases the first bus **7** and reads the instruction (set of data) from the RAM **12**. Then, the output unit **15** transfers the read instruction (set of data) to the CPU **6** using the first bus **7**. Subsequently, the CPU **6** executes the instruction transferred thereto or uses the data transferred thereto. Thus, in the information processing device **1**, even when only the first bus **7** is provided without providing the second bus **8**, it becomes possible to continue with the normal operation of the computer program.
[0059] Meanwhile, for explanatory purposes, instead of using the IC **2**, it is also possible to use, for example, an application specific integrated circuit/field programmable gate array (ASIC/FPGA). Thus, is possible to make changes in an arbitrary manner.
[0060] Moreover, the process functions implemented in the device are entirely or partially implemented by a CPU or by computer programs that are analyzed and executed by a CPU, or are implemented as hardware by wired logic.
[0061] According to an aspect, it becomes possible to continue with the normal operation of a computer program.
[0062] Although the invention has been described with respect to specific embodiments for a complete and clear disclosure, the appended claims are not to be thus limited but are to be construed as embodying all modifications and alternative constructions that may occur to one skilled in the art that fairly fall within the basic teaching herein set forth.

Claims

1. An information processing device having a processor configured to write a computer program, which is read from an external memory, in an internal memory and execute the computer program which is read from the internal memory, the information processing device comprising the processor configured to: calculate a parity value of an instruction or a set of data of the computer program read from the external memory, and write the instruction or the set of data and the parity value in the internal memory; determine, based on the parity value of the instruction or the set of data of the computer program read from the internal memory, presence or absence of an error in the instruction or the set of data of the computer program; and transfer, when the error has occurred, the instruction or the set of data of the computer program, which is read again from the external memory, to the first writing of the computer program and the second writing of the instruction or the set.
2. The information processing device according to claim 1, wherein the transferring includes: reloading, when the error has occurred, the instruction or the set of data of the computer program from the external memory, and transferring, when the error has occurred, to the first writing, the instruction or the set of data of the computer program that is again read by the reloading, and transferring, when the error has not occurred, to the first writing, the instruction or the set of data of the computer program that is read from the internal memory.

3. The information processing device according to claim 2, further including: a first bus that connects the transferring and the first writing unit to each other; and a second bus that connects the external memory and the transferring to each other, wherein when the error has occurred, the reloading again reads, from the external memory via the second bus, the instruction or the set of data of the computer program and transfers, to the transferring and the second writing, the instruction or the set of data of the computer program that is read again, when the error has occurred, the transferring transfers, to the first writing via the first bus, the instruction or the set of data of the computer program that is again read by the reloading, and when the error has not occurred, the transferring transfers, to the first writing via the first bus, the instruction or the set of data of the computer program that is read from the internal memory.
4. The information processing device according to claim 2, wherein the second writing calculates a parity value of the instruction or the set of data of the computer program that is again read by the reloading, and writes, in the internal memory, the instruction or the set of data of the computer program, which is again read, and the calculated parity value.
5. An information processing method implemented by an information processing device having a processor configured to write a computer program, which is read from an external memory, in an internal memory and executes the computer program which is read from the internal memory, the information processing method comprising: writing that includes calculating a parity value of an instruction or a set of data of the computer program read from the external memory, and writing the instruction or the set of data and the parity value in the internal memory; determining, based on the parity value of the instruction or the set of data of the computer program read from the internal memory, presence or absence of an error in the instruction or the set of data of the computer program; and transferring, when the error has occurred, the instruction or the set of data of the computer program, which is read again from the external memory, to the first writing of the computer program.